

Introduction

The aim of this project is to research and develop a Software-Defined Radio (SDR) system on an enterprise level Radio Frequency System-on-Chip (RFSoc) board, the RFX-8440 from BittWare.



Figure 1: BittWare RFX-8440 FPGA Card [1]

The project uses GNU Radio as a learning platform to understand the relevant theory behind modern communication systems. GNU Radio is ideal for this purpose as it is free and open-sourced, making it extremely accessible.

Methodology

The RFSoc is a specialized system-on-chip (SoC) developed by AMD, designed for RF signal processing. It includes dedicated resources such as four RF-ADC and RF-DAC channels, 16 GTY transceivers, and support for up to 100G Ethernet through two QSFP28 transceivers.

Objectives:

- Research SDR and the relevant communications theory.
- Gain experience with AMD RFSoc devices.
- Develop a functioning SDR system, using a loopback configuration, with a custom data generation (transmitter) and receiver IP cores.

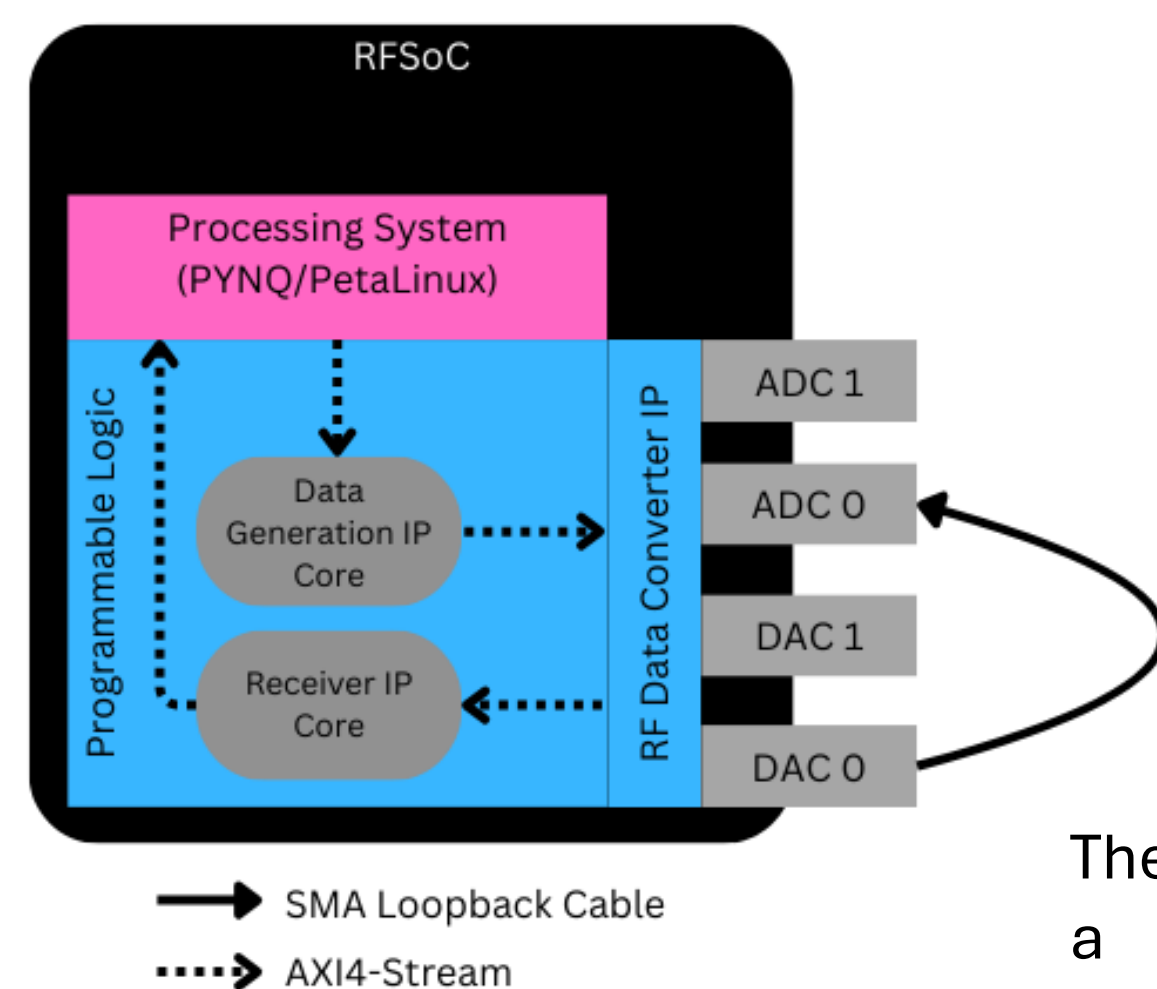


Figure 2: High level Block Diagram of Loopback system

Access to the RFX was limited, so to aid in development the Xilinx RFSoc 2x2 development board was used to aid in the hardware design, as this board is more accessible.

The RFX-8440 Design uses a reference design from BittWare which configures all aspects of the RFSoc.

Hardware Design

The Programmable Logic (PL) was designed using Vitis Model Composer to implement an Amplitude Modulation (AM) transmitter and receiver, which generates an AM sinusoid. This was chosen because the received signal should be easily identifiable.

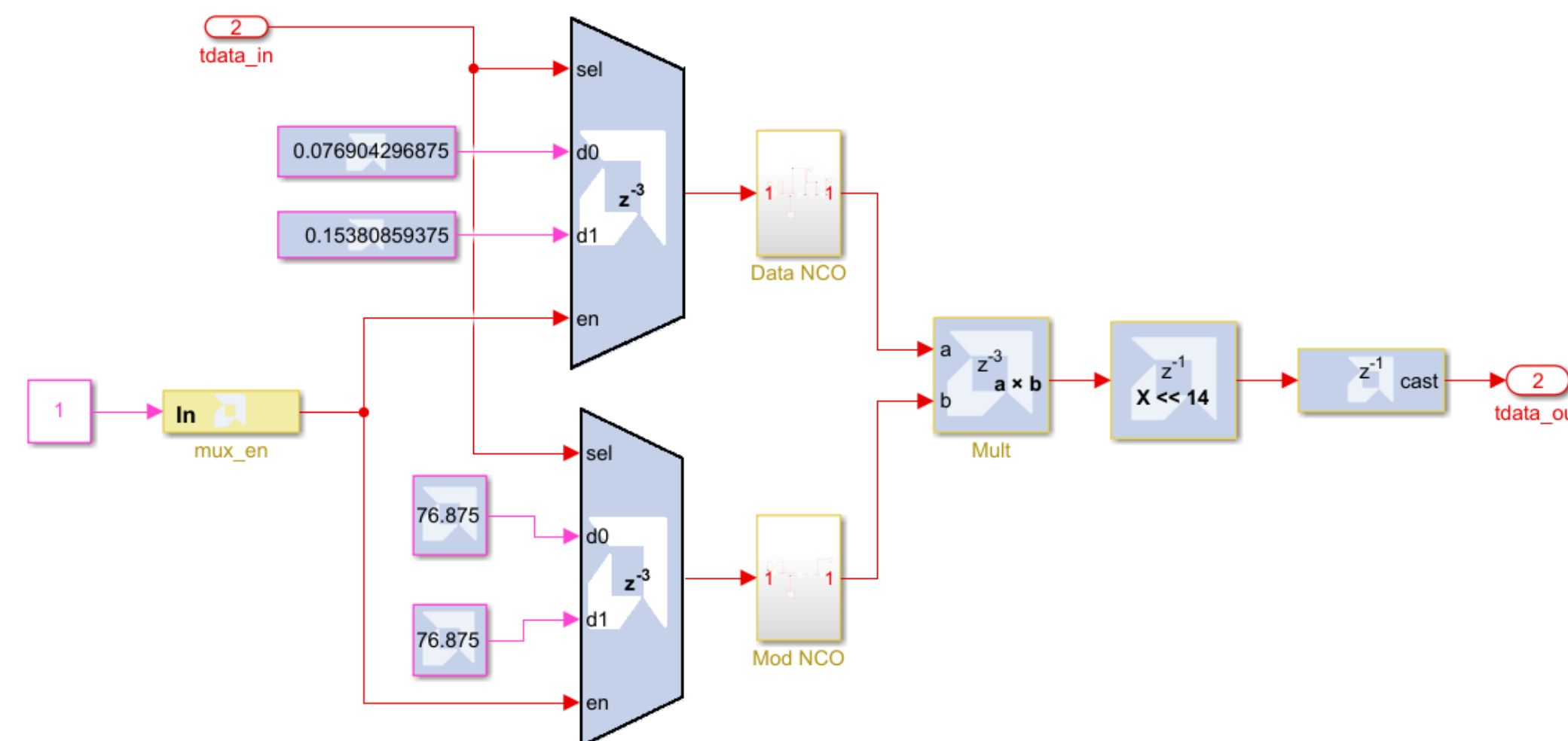


Figure 3: Transmitter Design in Model Composer

The signal is generated through two Numerically Controlled Oscillators (NCOs), which take a step size input to generate a different output frequency, dependent upon the sampling frequency.

Software Design

The Processing System (PS) design allows the interfacing between different parts of the hardware design. On the RFX-8440, this was done through PetaLinux, an embedded Linux OS image which runs on the PS. However, the RFSoc 2x2 uses PYNQ, a software which utilizes Python Application Programming Interfaces (APIs) for PS-PL communication and hardware configuration. For example, although the RF Data Converters (RFDCs) are preconfigured in hardware, they must also be initialised in software to ensure correct functionality. To communicate between the PS and PL, a Direct Memory Access (DMA) controller IP core is used to read and write DMA buffers from memory, which allows for high-speed data transfer between systems.

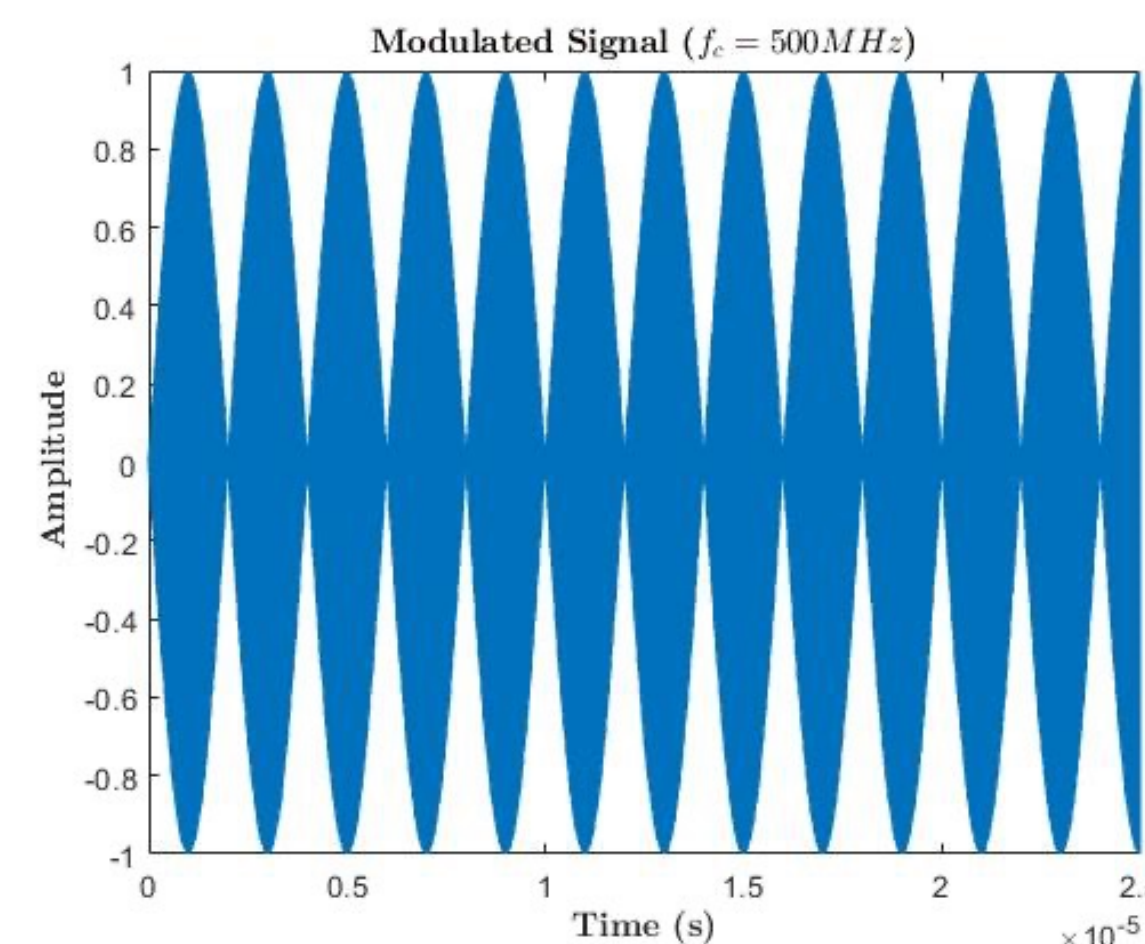


Figure 4: Modulated Transmitter Signal

In this case, PetaLinux also uses Python to initialise different functions on the PL. This was done through custom packages created by BittWare.

The generated signal has a base frequency of 100kHz, with a carrier frequency, f_c , of 500MHz.

Hardware Implementation

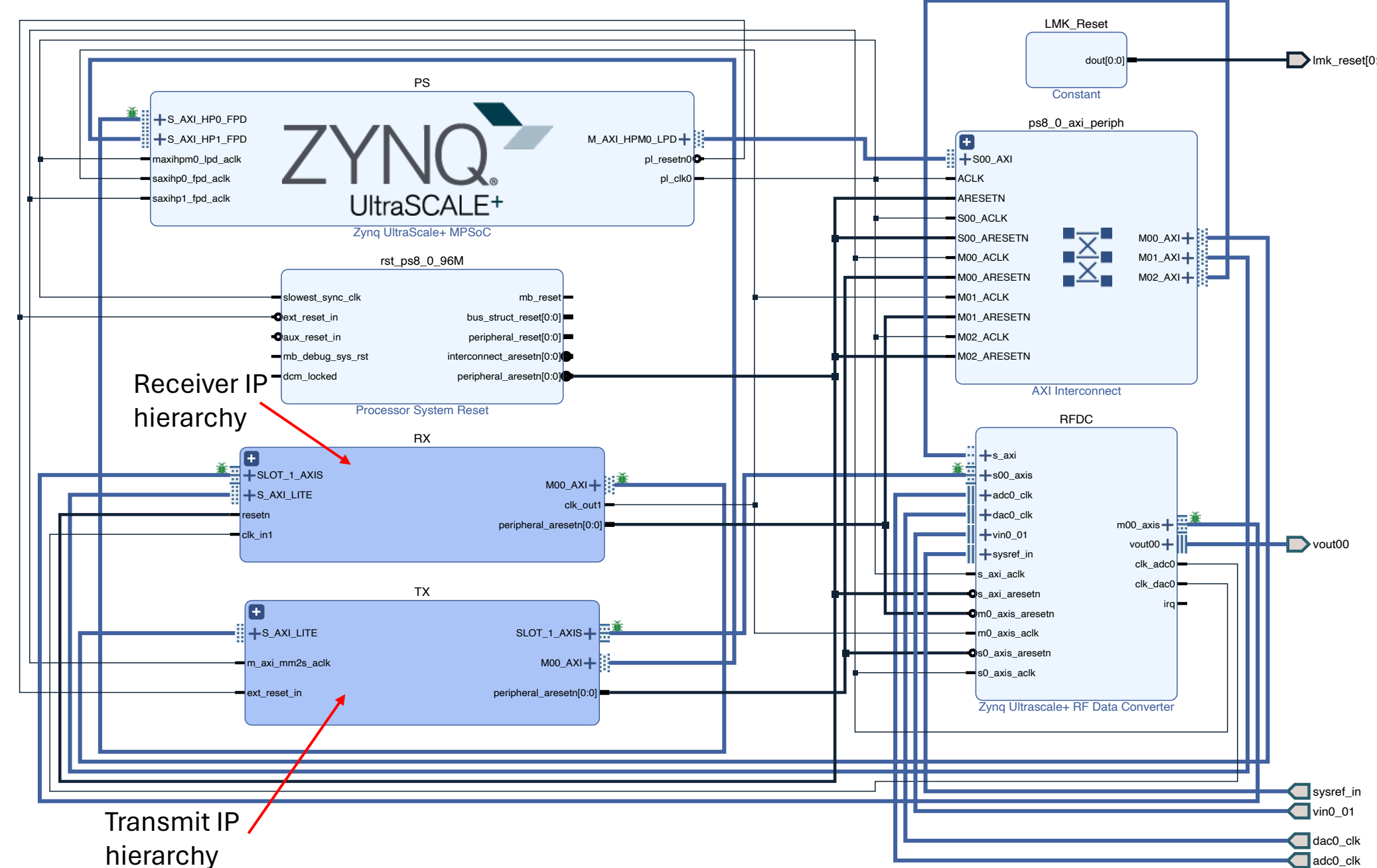


Figure 5: SDR System in Vivado IP Integrator on RFSoc 2x2

Model Composer designs can be exported as Hardware Description Language (HDL) code, from which the design can be added into Vivado's IP Integrator environment as a custom IP Core.

Resource	Utilisation	Available	Utilisation %
LUT	8006	425280	1.88
LUTRAM	1372	213600	0.64
FF	10754	850560	1.28
BRAM	16	1080	1.48
DSP	39	4272	0.91

Figure 6: Resource Allocation Table for 2x2 Design

Conclusions

- A transmitter and receiver were designed using Model Composer and implemented onto the RFSoc using Vivado, PYNQ and PetaLinux.
- The system functions effectively and consumes only a very small proportion of resources on the RFSoc device.
- This development work has demonstrated the design, testing, and implementation of an SDR system on the RFSoc.

Further Work

In future, further functionality could be added to the BittWare RFX-8440. The transceiver could be upgraded to feature a more complex design, exploring different modulation schemes to model more realistic communication systems. In addition, integrating GNU Radio into the design through a QSFP offload system would provide a robust and modular software design.

References

- [1] BittWare, a Molex Company, "RFX-8440 Framework Logic Guide." <https://www.bittware.com/products/rfx-8440a/>, 2024. Accessed: 19th March 2025.